

REMARKS ON INPUT/OUTPUT LINEARIZATION*

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Abstract

In the context of realization theory, conditions are given for the possibility of simulating a given discrete time system, using immersion and/or feedback, by linear or state-affine systems.

1. Introduction

A number of papers have been written recently on the topic of system simulation by simpler types of systems, like linear and state-affine systems. In particular, we shall be concerned here with the notion of *immersion* introduced by M.Fliess, and more specifically, in regard to *discrete-time* nonlinear systems (see [1,2,3]).

In this note we remark that the setup in [4], which deals with nonlinear realization, generalizes in an almost immediate manner to deal with questions of immersion, resulting in conceptually simple statements and proofs. The approach resulting from this generalization should be compared to that in [1,2,3], which relies on a theory for real-analytic (in fact, entire!) systems and bases all proofs on appropriate power series expansions.

2. Responses and Systems

We shall need a number of definitions. Except for the generalization from single responses to families of responses, these are precisely as in [4].

A set U (input-value set) and a finite-dimensional vector space Y (output-value set) over a fixed field k will be assumed fixed. A *response* is a mapping $f:U^* \rightarrow Y$, where U^* denotes the free monoid over U . A *family (of responses)* Φ is any set of responses, which we write in a parametrized way as $\Phi = \{f^\lambda, \lambda \in \Lambda\}$. The response f is of *type J*, where $J = \{\delta_0, \dots, \delta_m\}$ is a set of linearly independent functions $U \rightarrow k$ with $\delta_0 \equiv 1$, iff for each $t \geq 1$ there are (finitely many) vectors a_α in Y such that

$$f(\omega) = \sum \delta_\alpha(\omega) a_\alpha$$

where the summation runs over J^t and, if $\alpha = (\alpha_1, \dots, \alpha_t)$ and

$\omega = (u_1, \dots, u_t)$, then $\delta_\alpha(\omega)$ is the ('tensor') product

$\delta_{\alpha_1}(u_1) \cdots \delta_{\alpha_t}(u_t)$. The family Φ is of *type J* iff each f^λ in Φ is. The family Φ is *affine* iff $U = k^m$, Φ is of *affine type* $J_0 := \{\delta_0, \dots, \delta_m\}$ with $\delta_i =$ projection of k onto i -th factor, and the restrictions $f^\lambda|_{U^t}$ are all affine; Φ is *uniformly affine* if it is affine and the maps

$$f^\lambda_{|in}(\omega) := f^\lambda(\omega) - f^\lambda(\underline{0})$$

are all the same (i.e. independent of λ), say f . ($\underline{0}$ denotes here a zero input of length equal to $\text{length}(\omega)$) and f is a (time-invariant) linear i/o response, i.e., the restrictions of f to all U^t are linear, and $f(\underline{0}, \omega) = f(\omega)$, for any input sequence ω and any constantly 0 input sequence. Such families appear when dealing with affine systems (see below).

A *system* $\Sigma = (X, P, Q)$ consists of a set X and maps $P: XXU \rightarrow X$, $Q: X \rightarrow Y$. The system Σ is *state-affine* if X is a k -vector space, and Q as well as each map $P(\cdot, u)$, $u \in U$ are affine. It is *affine* if P and Q are affine maps (i.e., the system is basically *linear* in the usual sense, except that we do not a priori impose any equilibrium state requirements), and is of *type J* (J as above) if there exist affine maps P_i such that $P(x, u) = \sum \delta_i(u) P_i(x)$. Note that a *finite-dimensional* state-affine system is always of type J , for some (finite) J . We use the notation P^* for the recursive extension of P to input sequences: $P^*(x, \phi) := x$ (ϕ =empty sequence) and $P^*(x, \omega u) := P(P^*(x, \omega), u)$.

For any state $x \in X$, f^x is the response obtained when starting Σ at initial state x . The (*response*) *family* Φ_Σ of Σ is the family $\{f^x, x \in X\}$. When $\Phi \subseteq \Phi_\Sigma$ we say that Σ *realizes* the family Φ . Note that if Σ is (state-affine) of type J , then Φ_Σ is also of type J . And if Σ is affine, then Φ_Σ is also uniformly affine.

Let Σ and $\tilde{\Sigma}$ be two systems, with response families Φ and $\tilde{\Phi}$ respectively. The system Σ is *immersed* into the system $\tilde{\Sigma}$ iff $\Phi \subseteq \tilde{\Phi}$, i.e., $\tilde{\Sigma}$ realizes Φ . Another way of saying this is that there is a mapping $\phi: X \rightarrow \tilde{X}$ such that $\phi(x)$ and x give rise to the same response.

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(The definition used in [1,2,3] for analytic systems requires that there be an *analytic* such mapping. We prefer the more abstract definition, whenever the existence of an immersion will be concluded, the proof will in fact give a 'nice' immersion, depending on the class of systems in question: analytic, continuous, etc.)

Immersion is of course closely related to *morphisms* in the sense usual in system theory ($\phi(P(x,u)) = P(\phi(x),u)$ and $Q(x) = Q(\phi(x))$): a morphism is necessarily an immersion, and if Σ is observable the converse also holds. We shall not explicitly use the concept of morphism in this note, however.

Finally, we need the notion of the *observation space* L^Σ associated to Σ . This is the space of mappings from X into Y obtained in the following way. For each input sequence $\omega \in U^*$ (possibly empty) let $Q^\omega(x) = Q(P^*(x,\omega))$ = last output that results from applying input sequence ω to initial state x . The set of all maps $[X \rightarrow Y]$ is endowed with a natural k -linear structure (pointwise operations); then L^Σ is defined as the linear span of $\{Q^\omega, \omega \in U^*\}$.

An important observation is that L^Σ is *finite-dimensional* when Σ is a finite-dimensional *state-affine* system, since in that particular situation all the generators Q^ω are affine maps on a finite-dimensional space. Another relevant fact is:

Proposition 1. If Σ is immersed in $\tilde{\Sigma}$ then there is a surjective mapping from $L^{\tilde{\Sigma}}$ onto L^Σ . In particular, if Σ is immersed in a finite-dimensional state-affine system then L^Σ is finite-dimensional.

Proof. Let $\phi^*: [X \rightarrow Y] \rightarrow [X \rightarrow Y]$ be the dual operator defined by the composition $\phi^*(T) = T \circ \phi$. This is a linear map, and its restriction to $L^{\tilde{\Sigma}}$ maps into (and in fact, onto) L^Σ .

3. Canonical Realizations

Let $\Phi = \{f^\lambda, \lambda \in \Lambda\}$ be a fixed family as in the previous section. We associate to Φ the "Nerode space" L_Φ defined as follows. For each f^λ and each $\omega \in U^*$ let f^λ_ω be the response defined by $f^\lambda_\omega(\nu) = f^\lambda(\omega\nu)$ (i.e., 'preset' by input ω). The set of all responses is naturally a k -vector space under pointwise operations; L_Φ is then defined as the *affine hull* of

$$\{f^\lambda_\omega, \omega \in U^*, \lambda \in \Lambda\}.$$

(We could use here instead the *linear span*; basically the same theory would result, except that state-linear systems would appear at various points instead of state-affine ones.)

If Σ is a realization of Φ , there is a natural duality between L_Φ

and the observation space L^Σ defined earlier. Consider the pairing

$$(\cdot, \cdot): L_\Phi \times L^\Sigma \rightarrow k$$

obtained by bilinearly extending the operation on generators

$$(f^\lambda_\omega, Q^\nu) := f^\lambda(\omega\nu).$$

It is necessary to see that this extension is well defined. Assume then that

$$\begin{aligned} \Sigma \alpha_{\omega\lambda} f^\lambda_\omega &= \Sigma \beta_{\omega\lambda} f^\lambda_\omega \quad \text{and} \\ \Sigma \gamma_\nu Q^\nu &= \Sigma \delta_\nu Q^\nu \end{aligned}$$

where the first pair of sums are over a (finite) set of pairs (ω, λ) and the second over some (finite) set of inputs ν . Then, $\Sigma \gamma_\nu \Sigma \alpha_{\omega\lambda} f^\lambda_\omega = \Sigma \gamma_\nu \Sigma \beta_{\omega\lambda} f^\lambda_\omega = \Sigma \gamma_\nu \Sigma \beta_{\omega\lambda} f^\lambda_\omega = \Sigma \gamma_\nu \Sigma \beta_{\omega\lambda} f^\lambda(\omega\nu) = \Sigma \beta_{\omega\lambda} \Sigma \gamma_\nu f^\lambda(\omega\nu)$, and it is enough to prove that $\Sigma \gamma_\nu f^\lambda(\omega\nu) = \Sigma \delta_\nu f^\lambda(\omega\nu)$ for each fixed (ω, λ) . But realizability of Φ by Σ means that there is an $x \in X$ such that $f^x = f^\lambda$. Let $z := P^*(x, \omega)$ for this x and the given ω . Thus, $f^\lambda(\omega\nu) = Q^\nu(z)$ for each ν , so $\Sigma \gamma_\nu f^\lambda(\omega\nu) = (\Sigma \gamma_\nu Q^\nu)(z) = (\Sigma \delta_\nu Q^\nu)(z) = \Sigma \delta_\nu f^\lambda(\omega\nu)$, as desired. Further, this pairing is nondegenerate in the first term, i.e., $(\Sigma \alpha_{\omega\lambda} f^\lambda_\omega, \Sigma \gamma_\nu Q^\nu) = 0$ for all $\Sigma \gamma_\nu Q^\nu$ implies that $\Sigma \alpha_{\omega\lambda} f^\lambda_\omega = 0$. It will also be nondegenerate in the second term iff the condition " $\Sigma \gamma_\nu Q^\nu = 0$ on all elements of the form $P^*(x, \omega)$ such that ω is an input sequence and $f^x \in \Phi$ " implies $\Sigma \gamma_\nu Q^\nu = 0$. This happens in various cases of interest, for instance if there is some kind of 'span-reachability' for the system as in [4], and in any case certainly when $\Phi = \Phi_\Sigma$. We conclude that:

Proposition 2. If $\Phi = \Phi_\Sigma$, then L_Φ is finite dimensional if and only if L^Σ is finite dimensional.

We now show how to construct a 'canonical' realization Σ_Φ of any given family Φ . This realization will have the property that it is a finite dimensional state-affine system whenever L_Φ is finite dimensional, and is affine whenever Φ is also uniformly affine.

The state space of Σ_Φ is by definition L_Φ , seen as a vector space. The maps P and Q are defined as follows. For each $u \in U$, let $P(\cdot, u)$ be the (well-defined!) extension of $P(f^\lambda_\omega, u) = f^\lambda_{\omega u}$, and let Q be the extension of the evaluation $Q(f^\lambda_\omega) = f^\lambda(\omega)$. This system is *state-affine*, by construction. Applying proposition 2 to $\Phi = \Phi_\Sigma$, and using proposition 1, we then conclude:

Theorem 1. Σ can be immersed in a finite dimensional state-

affine system if and only if its observation space is finite dimensional.

This result can be made more precise in a number of ways. Assume for instance that Φ is of type J. Then Σ_Φ is also of type J. Indeed, write, for any $\omega \in U^*$ and $u \in U$,

$$f^\lambda(\omega u) = \sum \delta_\alpha(\omega) \delta(u) a_\alpha^\lambda.$$

Then, $P_i(x, u) = \sum \delta(u) P_i(x)$, where P_i is defined on generators by the formula

$$P_i(f_\omega^\lambda) = \sum \delta_\alpha(\omega) a_\alpha^\lambda.$$

The P_i are well-defined (use linear independence of the $\delta(u)$ to verify this) and are affine. We then have:

Theorem 2. The system Σ can be immersed into a [finite dimensional] system of type J if and only if Φ_Σ is of type J [and L^Σ is finite dimensional].

Now assume that Φ is uniformly affine, and consider its canonical realization Σ_Φ . We claim that this is an affine system. Since Σ_Φ has affine type J_0 , we only need to establish that all P_i , $i = 1, \dots, m$ are constant. Note that we may write

$$(1) \quad f^\lambda(\omega u) = \sum u_i P_i(f_\omega^\lambda) + P_0(f_\omega^\lambda),$$

$$(2) \quad f(\omega u) = f^\lambda(\omega u) - f^\lambda(00),$$

where f is linear and independent of λ , and satisfies $f(0, u) = f(u)$ as in the definition. It follows from (2) that $f_\omega^\lambda(u) - f_\omega^\lambda(0) = f(\omega u) - f(\omega 0) = f(0, u) = f(u)$, while from (1) this expression equals $\sum u_i P_i(f_\omega^\lambda)$. Comparing coefficients of each u_i , the desired constancy is deduced. Thus we conclude:

Theorem 3. The system Σ can be immersed in a [finite dimensional] linear system if and only if Φ_Σ is uniformly affine [and L^Σ is finite dimensional].

This is, with somewhat different terminology, the characterization given in [1,2,3] for immersions of analytic Σ into finite dimensional systems. The only difference is that the finite dimensionality condition given there is not in terms of L^Σ —which is natural in the general context treated here—but in terms of L_0^Σ , defined as follows.

Assume that $U = k^m$. The 0-input observation space L_0^Σ is the subspace of L^Σ generated by the Q^ω with $\omega = a$ a sequence

of the type 0 (all zero). In case Φ_Σ is a uniformly affine family, for any observable we have that

$$Q^\omega(x) = f^x(\omega) = f^x(0) + f(\omega) = Q^0(x) + a_\omega,$$

where a_ω is a constant (vector in Y) depending only on ω . That is, L^Σ is in that case included in the sum of L_0^Σ and a finite dimensional space (constant functions), so that

$$L^\Sigma \text{ is finite dimensional iff } L_0^\Sigma \text{ is.}$$

4. Feedback

We give now some results on linearization under feedback. These results are basically just restatements of those in [1], proved in slightly more generality using a more abstract approach. Further work along these lines is in progress.

Fix a parameter set Λ ; all families of responses will be parametrized by this set. A system-like family is one for which the following property holds: for each $\lambda \in \Lambda$ and $u \in U$ there is a $\mu \in \Lambda$ such that $f^\lambda(\omega u) = f^\mu(\omega)$ for all $\omega \in U^*$. Such families appear when considering the families Φ_Σ .

An i/o family will mean a family $\Gamma = \{g^\lambda, \lambda \in \Lambda\}$ of maps $g^\lambda: U^* \rightarrow U^*$ each of which is length preserving. Given a family of responses Φ , and a Γ like this, we define the new family of responses $\tilde{\Phi} = \Phi_\Gamma$ via the compositions $\tilde{f}^\lambda := f^\lambda \circ g^\lambda$, and denote $\tilde{\Phi} \leq \Phi$ if $\tilde{\Phi}$ can be obtained in this way from Φ . We also denote $\tilde{\Phi} \sim \Phi$ if both $\tilde{\Phi} \leq \Phi$ and $\Phi \leq \tilde{\Phi}$ hold. Consider the degree of the family Φ :

$$d = d(\Phi) := \max\{t \text{ s.t. } f^\lambda|_{U^j} \text{ constant if } j \leq t \text{ and } \lambda \in \Lambda\},$$

and assume that $d < \infty$ for the systems here considered. For $j \leq d$, $v_j(\lambda)$ denotes the common value of the restriction of f^λ to U^j . Since the elements of Γ preserve length, it is clear that if $v_j(\lambda)$ is like this and $\tilde{\Phi} = \Phi_\Gamma$ then also

$$(3) \quad \tilde{f}^\lambda|_{U^j} = v_j(\lambda), \text{ and so } (4) \quad \tilde{d} \geq d.$$

(In particular, $\tilde{d} = d$ when $\tilde{\Phi} \sim \Phi$.) Further, for any $\nu \in U^+$, any $\lambda \in \Lambda$, and any $\omega_1 \in U^d$, $\tilde{f}^\lambda(\nu \omega_1) = f^\lambda(g^\lambda(\nu, \omega_1))$, for some ω of length d . But Φ being system-like implies that $f^\lambda(g^\lambda(\nu, \omega_1)) = f^\mu(\omega)$ for some μ independent of ω , and the definition of d implies that this last expression equals $f^\mu(\omega_2)$ for every other ω_2 of length d , so that, for all such ω_1 ,

$$(5) \quad \tilde{f}^\lambda(\nu \omega_1) = f^\lambda(g^\lambda(\nu, \omega_2)).$$

As in [1], we only study here feedback immersions into a particular (and very restrictive!) class of linear systems. Call a family Φ special iff it has finite degree and for some $d \leq \text{deg}(\Phi)$ there is a linear transformation $D: U \rightarrow Y$ such that

$$f^\lambda(u0^d) = \sum_j a_j v_j(\lambda) + Du$$

for all $u \in U$ (sum is over $j=0, \dots, d$). Such a family can be immersed into a uniformly affine finite dimensional system, as done for the analogous situation in [1]. For example, the linear system with state space $X := Y^{d+1}$ and equations (in Y -block form) $x_i' = x_{i+1}$, $i=0, \dots, d-1$, and $x_d' = \sum_j a_j x_j + Du$, and output $y = x_0$, realizes such a family with initial states $(v_0(\lambda), \dots, v_d(\lambda))$.

Note that, conversely, if $\Sigma = (A, B, C)$ is a linear system, then Φ_Σ is special iff there is a d such that $CA^j B = 0$ for all $j = 0, \dots, d-1$ and CA^{d+1} is a linear combination of the CA^j , $j \leq d$. A somewhat more general definition of 'special' could be given, in the style of [1], where the property is required of each output channel separately; the results are analogous but notations become somewhat more involved.

Let d be the degree of Φ . Consider the following property for Φ and Γ :

(6) For some linear D , $f^\lambda(g^\lambda(u), 0^d) = \sum_j a_j f^\lambda(0^j) + Du$ for all $\lambda \in \Lambda$.

Proposition 3. If there is an i/o family Γ such that (6) is satisfied then Φ_Γ is special.

Proof. Let $\tilde{\Phi} = \Phi_\Gamma$. By formula (5), $\tilde{f}^\lambda(u, 0^d) = f^\lambda(g^\lambda(u), 0^d)$,

which is as in (6). By (3), $f^\lambda(0^j) = v_j(\lambda)$; thus $\tilde{\Phi}$ is special.

Let $\Phi = \Phi_\Sigma$, for a given system Σ . We say that the i/o family Γ is *feedback-like* for Φ iff there exists a map $K: XXU \rightarrow U$ such that $K(x, \cdot)$ is invertible for each fixed x , and each g^λ is defined as usual by the outputs of the system obtained from $x' = P(x, K(x, u))$ and $y = K(x, u)$. If this K is of the form $F(x) + G(x)u$, with $G(x)$ invertible, we say the feedback is *affine*. Note that the invertibility condition insures that Φ_Γ can be obtained, conversely, from [affine] feedback applied to Φ . (Somewhat less than invertibility is needed for the results to follow; also, a more general, input/output, definition can be given of feedback-like Γ .)

Corollary to Proposition 3. There is a feedback-like [resp., affine] Γ such that Φ_Γ is special iff condition (6) is satisfied with some such Γ .

Proof. Sufficiency is just Proposition 3. Conversely, if there is such a Γ then $\tilde{d} = d$ for the degrees of $\tilde{\Phi} = \Phi_\Gamma$ and Φ respectively (from (4)). Since $\tilde{\Phi}$ is special, formulas (3) and (5) give that (6) must hold.

Assume that there is an affine Γ with $\tilde{\Phi} = \Phi_\Gamma$ uniformly affine, and let $\Delta = \{\delta^\lambda\}$ be the 'inverse' of Γ , computed from the generating feedback H that is the pointwise inverse of K . Since $\Phi = \tilde{\Phi}_\Delta$, we may apply formula (5) to this pair. Thus

$$f^\lambda(v0^d) = \tilde{f}^\lambda(\delta^\lambda(v), 0^d) = s(\lambda) + R(\lambda)v,$$

for some s, R (because $\tilde{f}^\lambda(u, 0^d)$ and δ^λ are each affine). Assume now that $f^\lambda(v0^d)$ has this affine form. Then the above corollary says that an affine Γ exists making Φ_Γ special iff there are F, G such that, (substituting the form $g^\lambda(u) = F(\lambda) + G(\lambda)u$ into the left hand side of (6)),

$$R(\lambda)G(\lambda) = D = \text{constant, and} \\ s(\lambda) + R(\lambda)F(\lambda) = \sum_j a_j v_j(\lambda).$$

Thus one recovers the conditions in [1], since clearly $s(\lambda) = f^\lambda(0^{d+1})$ and $v_j(\lambda) = f^\lambda(0^j)$.

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